

Sunilkumar Pathipati

Lead Engineer | Python | AI | Software Test Framework Development | AUTOSAR

Bengaluru, Karnataka, India

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PROFESSIONAL SUMMARY

Lead Engineer with 9+ years of experience in automotive software engineering, embedded systems, artificial intelligence, and application development. Expertise in Python, C, AUTOSAR, machine learning, and Agile development practices. Experienced in developing safety-critical automotive solutions, including anti-lock braking systems for eBikes and two-wheelers, with strong exposure to ISO 26262 functional safety environments. Skilled in Python-based automation frameworks, local LLM tooling, intelligent testing platforms, software integration, verification and validation, release management, and engineering productivity improvement.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, Embedded C, Bash, MATLAB
- **AI & Machine Learning:** Artificial Intelligence, Machine Learning, LLMs, llama.cpp, Data Analytics
- **Automotive:** AUTOSAR, ISO 26262, Android Infotainment, ABS Systems, Embedded Systems
- **Tools & Technologies:** Git, GitHub, GitHub Actions, JFrog Artifactory, Linux, Simulink
- **Core Competencies:** Test Automation Frameworks, Software Validation, System Integration, Release Management
- **Methodologies:** Agile Development, CI/CD Workflows, Software Development Life Cycle, Verification & Validation
- **Languages:** English, Telugu, Hindi, Kannada, Tamil

PROFESSIONAL EXPERIENCE

- **Lead Engineer** September 2025 – Present
BlueBinaries Engineering and Solutions Pvt Ltd
Bengaluru, Karnataka, India
 - Client: Polestar
 - Developing Python-based automation frameworks and local LLM solutions using llama.cpp for automotive infotainment testing and validation.
 - Building AI-powered tools for test generation, log analysis, defect investigation, knowledge retrieval, and engineering productivity.
 - Designing and maintaining GitHub Actions workflows, Git-based development processes, and automation utilities to streamline testing, build, and release activities.
 - Leveraging Python, Bash scripting, and JFrog Artifactory to improve software quality, engineering efficiency, and validation workflows.
 - Supporting secure on-premises AI tooling and local model deployment for automotive software testing environments.
- **Lead Engineer** December 2024 – August 2025
Bosch Global Software Technologies
Bengaluru, Karnataka, India
 - Led automotive software development and validation activities for safety-critical embedded systems.
 - Managed software integration, release management, verification, and validation activities.
 - Collaborated with cross-functional teams to improve software quality, reliability, and delivery efficiency.
- **Lead Engineer** July 2024 – November 2024
Bosch Global Software Technologies
Reutlingen, Baden-Wurttemberg, Germany
 - Led automation initiatives and engineering workflow improvements across global automotive software teams.
 - Supported application development, testing, integration, and process optimization activities.
 - Worked with international teams to improve development efficiency and knowledge transfer.

- Lead Engineer** January 2024 – June 2024
Bosch Global Software Technologies *Bengaluru, Karnataka, India*
 - Contributed to algorithm development, application development, and software release activities for automotive systems.
 - Supported ISO 26262-compliant development and validation processes for safety-critical software.
- Specialist** January 2023 – December 2023
Bosch Global Software Technologies *Bengaluru, Karnataka, India*
 - Developed and validated automotive software components for eBike and two-wheeler braking systems.
 - Designed automation solutions and test frameworks to improve validation efficiency and software quality.
- Senior Software Engineer** June 2022 – January 2023
Bosch Global Software Technologies *Bengaluru, Karnataka, India*
 - Worked on embedded software development, testing, debugging, and integration for automotive applications.
 - Supported MATLAB/Simulink-based algorithm development and validation activities.
- Senior Software Engineer** December 2021 – June 2022
Bosch Global Software Technologies *Reutlingen, Baden-Wurttemberg, Germany*
 - Supported global automotive software development and testing activities in Germany.
 - Collaborated with international engineering teams on validation, automation, and software quality improvements.
- Software Engineer** July 2019 – December 2021
Bosch Global Software Technologies *Bengaluru, Karnataka, India*
 - Developed embedded automotive software and supported testing activities for safety-critical systems.
 - Worked on software verification, validation, debugging, and release support.
- Software Engineer** March 2017 – July 2019
HCL Technologies *Bengaluru, Karnataka, India*
 - Developed and tested embedded software solutions for automotive clients.
 - Supported system integration, validation, debugging, and software testing activities.
 - Worked with Linux-based development environments and automotive software tools.

EDUCATION

- Bachelor of Technology in Electronics and Communications Engineering** 2012 – 2016
Ramireddy Subbaramireddy Engineering College *Andhra Pradesh, India*

CERTIFICATIONS

- Certified Data Scientist
- Data Science Foundation
- Deep Learning with Keras and TensorFlow